Math 421

Wednesday, October 15

Announcements

- Exam 1
  - **Today** 5:45-7:15 pm
  - Van Vleck B130
- Drop-in Hours
  - **Today** 12-1 pm **VV B205**

Exam 1 Review: Proof Templates and Practice

Chapter 0

Proof Template – Indirect Proofs

When using an indirect proof to prove a statement: “If [P], then [Q]”

- It is a **good idea** to start a proof by contrapositive with
  - “We prove the contrapositive: If [not Q], then [not P]. Assume [not Q]…”

- It is a **good idea** to start a proof by contradiction with either
  - “We argue by contradiction. Suppose [P] and [not Q]”

    OR
  - “Suppose for a contradiction that [P] and [not Q].”

Practice – Prove Using Contrapositive

Theorem. For every integer n, if n^3 is odd, then n is odd.

Bonus Activity: Prove using contradiction.

Proof Template for Sets

1. To show that $T \subseteq S$:
   I. "Fix $t \in T$."
   II. Show that $t \in S$.
   III. "Since $t \in T$ was arbitrary, $T \subseteq S$."
2. To show that $T = S$:
   - Show that $T \subseteq S$.
   - Show that $S \subseteq T$.

Practice

Activity: Let $A = \{4k + 1 : k \in \mathbb{Z}\}$ and $B = \{4k - 3 : k \in \mathbb{Z}\}$. Prove that $A = B$.

Chapter 1

Mathematical Induction

Basic Template:

- "We argue by induction."
- "Base Case: If $n = 1$, then..."
  - Prove true for $n = 1$.
- "Induction Step: Assume [the statement] is true for some $n \in \mathbb{N}$."
  - Prove true for $n + 1$.

Practice – Prove Using Induction

Theorem. If $n \in \mathbb{N}$, then $\sum_{i=1}^{n} \frac{1}{i(i+1)} = 1 - \frac{1}{n+1}$.

Bonus Activity: Prove in a different way. (Hint: Break $\frac{1}{i(i+1)}$ into a difference)

Absolute Value & Distance

Definition. The **absolute value function** $|\cdot| : \mathbb{R} \to \mathbb{R}$ is defined by

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Recall: $|a - b|$ is the distance between $a, b \in \mathbb{R}$.

Triangle Inequality: $|a + b| \leq |a| + |b|$.

Proof Technique: Argue by cases or use the triangle inequality .

Rational/Irrational Numbers

The rational numbers $\mathbb{Q}$ are the numbers of the form $\frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$.

A real number is **irrational** if it is not rational.

Proof Technique: To show that a number is irrational, usually use a proof by contradiction.

Practice – Prove the Theorem

Theorem. If $x \geq 0$ is irrational, then $\sqrt{x}$ is irrational.

Proof Template for sup/inf

1. To show that y is an upper bound for $S \subseteq \mathbb{R}$:
   I. "Fix $s \in S$."
   II. Show that $s \leq y$.
   III. "Since $s \in S$ was arbitrary, y is an upper bound for S.
2. To show that $y = \sup S$:
   - Show that y is an upper bound for S. Hence $\sup S \leq y$.
   - Show that if u is an upper bound for S, then $y \leq u$. Hence y is the least upper bound for S.

Activity: Write a version of the above for lower bounds and the infimum

Practice – Prove the Theorem

Theorem. Let $S \subseteq \mathbb{R}$ be a non-empty bounded set and let

$$-S = \{-x : x \in S\}.$$

Then:

1. y is an upper bound for S if and only if $-y$ is a lower bound for $-S$.
2. $\inf(-\mathrm{S}) = -\sup S$.

Chapter 2

Proof Template for explicit limits

Given an explicit sequence, like $s_n = \frac{1}{n}$, to show that $\lim s_n = s$ using only the definition:

I. "Fix $\epsilon > 0$."
II. "Let $N = (something)$." [this requires scratch work]
III. "Assume $n > N$"
IV. Show that $|s_n - s| < \epsilon$.
V. "Since $\epsilon > 0$ was arbitrary, $\lim s_n = s$."

Practice

Activity: Prove, **using only the definition of the limit**, that

$$\lim \sqrt{n^3 - 1} - n^{3/2} = 0.$$

Common Tricks

If your assumptions include the fact that $\lim s_n = s$, your proof will probably have a sentence of the form

- Since $\lim s_n = s$, there exists N such that: if $n > N$, then $|s_n - s| < \boxed{}$

If your assumptions include the fact that $\lim s_n = s$ and $\lim t_n = t$, your proof will probably have sentences of the form

- Since $\lim s_n = s$, there exists N_1 such that: if $n > N_1$, then $|s_n - s| < \boxed{}$
- Since $\lim t_n = t$, there exists N_2 such that: if $n > N_2$, then $|t_n - t| < \boxed{}$
- Let $N = \max(N_1, N_2)$

Practice – Use only the definition of the limit

Theorem. If $\lim s_n = s$ and $\lim t_n = t$, then

$$\lim\,(2s_n + 3t_n) = 2s + 3t.$$